

Pictures and Figures:
Golding, Halliday, and the Stratum Closest to Being

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Abstract

This paper advances a convergence argument across phenomenological linguistics, narrative enactment, and embodied cognitive science. M. A. K. Halliday's Systemic Functional Grammar describes a stratum of meaning — the content plane of figures — that precedes grammaticalization: experience organised as configurations of process, participants, and circumstances before any lexicogrammatical structure captures it. This stratum is structurally identical to what William Golding's Neanderthal protagonists enact as “picture-thinking,” and to what enactivist cognitive science identifies as the sensorimotor coupling through which organisms constitute meaning prior to representation. Clark (2009) concludes that what previous analysis read as communicative deficit in Golding's Neanderthals may be richness at a level the grammaticalized system cannot register — a conclusion stratification theory reaches independently. Golding's prose enacts the content plane directly: transitivity suppressed, causality unencoded, likeness as figural identity before predicated comparison. Mapping Halliday's stratification model onto a companion paper's (a1)–(a2)–(b ↔ c) framework gives the argument structural precision: the history of representation becomes a history of progressive grammaticalization, AI represents the completion of that history by operating without any figural ground, and the a3 condition is the recovery of access to the content plane. A further section distinguishes the lateral a2 convulsion from the outward a1-touching convulsion, arguing that discriminating between them is the central practice of a3 existence. A final section identifies aging as a second and universal forcing condition, reading the gerontolinguistic pattern of preserved transformed gist in healthy aging as empirical evidence of outward movement through the strata.

Keywords: content plane, embodied cognition, stratification, artificial intelligence, aging, gist

1. Background: The Haft and the Eye

This paper develops an argument that begins in a companion piece, “The Haft and the Eye: Being, Representation, and the AI Epoch” (McKnight, under review). What follows presupposes that argument's framework and extends it into a territory the companion paper identifies but does not enter. A condensed account of the framework is offered here for readers approaching this paper independently.

Knowledge is built through negation: we understand things relationally, not in themselves. A cat is a cat because it is not a dog, not a table, not a sound. Every concept exists only within a web of differential relationships. If all knowledge is negational and relational, there must be something a priori that gets negated in the first place — a primordial, undifferentiated whole that exists before conceptual division. Following Heidegger (1993), the companion paper calls this Being (a1). The process of negation produces individual concepts — beings from Being — through a looping system of differential relationships designated (b) ↔ (c). Being is left behind, and a gap opens

between beings and Being. As Bissell and Levenstein (2011) demonstrate diagrammatically in their comparative study of Heidegger, Wittgenstein, and Nāgārjuna, new concepts progress in the opposite direction from Being: the intensification of the (b) ↔ (c) loop moves laterally within the encoded system, never back toward the primordial ground. Our concept of Being — (a2) — is a placeholder produced by the system, not the thing itself.

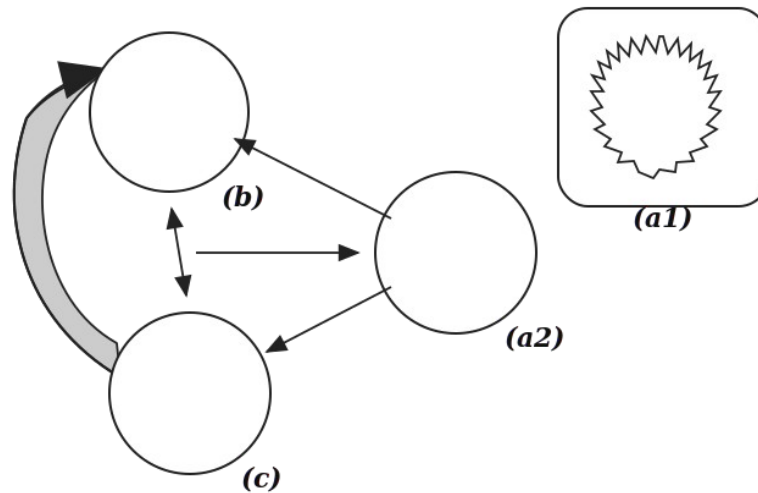


Fig. 1. Diagram illustrating the structural relationship between primordial being (a1), the separated negation state (a2), and the dynamic interplay of forces (b, c). Adapted from Bissell and Levenstein (2011).

William Golding's novel *The Inheritors* (1955) provides the companion paper's narrative foundation. Golding faced an impossible literary task: express what it is like to experience without conceptual language. His Neanderthal protagonists — Lok, Fa, and their small group — operate through what the paper calls picture-thinking: immediate, holistic, non-relational images that present experience as a unified whole. What they see is what it is, not what it is not. A log is bridge, canoe, kindling — not as multiple interpretations of a single object but as immediate encounters with what-is-there. There is no seeing-as, only seeing. The Neanderthals live close to (a1): absorbed, embodied, unified. They are good with blades. They have no hafts — no capacity to design across distance, to impose conceptual form on material, to think about thinking.

When *Homo sapiens* arrive with both practice and representation — both blade and haft — the displacement is not violent conquest but cognitive asymmetry. The novel's closing image shows Tuami, the craftsman, sailing away from the receding Neanderthal shoreline, unable to see whether the line of darkness has an ending, carving a haft he cannot stop shaping. What Tuami cannot see is that this capacity for designing hafts will eventually be perfected by something that has no hands at all: artificial intelligence.

The companion paper traces an epochal arc from pre-Socratic immediacy through Platonic idealism, Cartesian modernity, and Heideggerian Gestell (Heidegger 1993) to artificial intelligence, arguing that each epoch represents deeper immersion in the (b) ↔ (c) loop and further distance from (a1). AI represents the completion, not merely the intensification, of this arc: the representational project achieving such perfection that it no longer requires human hands to execute it. Where Neanderthals had practice alone (all blade, no haft) and were displaced by beings who brought both, AI inverts the structure: it brings representation alone, all haft, no hand. The work being automated is precisely the (b) ↔ (c) work — analysis, synthesis, abstract problem-solving, symbol manipulation. What resists automation requires embodiment: the capacity for absorbed, immediate, practice-based engagement with a world that pushes back.

This forced displacement may accomplish what philosophy argued but culture refused: a return to practice as primary. The companion paper calls the condition that results a3 — practice permanently inflected by having passed through full representational consciousness. Not regression to Neanderthal immediacy, which is not available to beings who have held the haft and built civilizations of hafts. Not a synthesis. A third condition, without prior name, in which the (b) ↔ (c) apparatus remains available but is no longer primary — in which what was always the ground of representation becomes accessible again as ground.

Central to the companion paper's argument — and the origin of this one — is a passage in chapter 10 of *The Inheritors*. Lok, confronting the incomprehensible Homo sapiens for the first time, undergoes what Golding calls a convulsion of the understanding: he discovers “like.” He had used likeness all his life without being aware of it — fungi on a tree were ears, not as metaphor but as figural identity, the similarity lived rather than observed. In this moment, likeness becomes visible as a tool: practice becoming aware of what it was already doing. The convulsion is not the creation of analogy but its first enunciation — the pre-predicative as-structure (Heidegger 1962: §32), always already operative in Lok's immediate practice, achieving explicit form for the first time. This passage is where the companion paper's argument reaches its philosophical center, and where this paper begins.

2. An Independent Convergence

The argument developed in “The Haft and the Eye” draws on William Golding's *The Inheritors* as narrative evidence for a philosophical claim: that Neanderthal consciousness operates through “picture-thinking” — immediate, holistic, non-relational images that present experience as a unified whole rather than as a set of conceptually mediated propositions. What the paper does not yet name is that this description has a precise structural counterpart in an entirely different disciplinary tradition — one that arrived at the same description not through a novelist's problem but through a linguist's.

M. A. K. Halliday's Systemic Functional Grammar describes a stratum of meaning that precedes grammaticalization: the content plane, within which experience is organized as figures — configurations of participants, processes, and circumstances perceived as unified wholes before any lexicogrammatical structure captures them. A figure is not a proposition about the world. It is the world as it presents itself in immediate experience: someone doing something, somewhere, somehow — but not yet encoded as subject-verb-object, not yet decomposed into the syntactic and lexical apparatus that would turn it into a clause.

Golding arrived at pictures through a novelist's problem — how to represent a consciousness without abstract language. Halliday arrived at figures through a linguist's problem — how to describe the stratum of experience that language encodes but does not create. Neither was working from the other. And yet they are describing the same thing from different directions. The convergence is itself a Lok discovery: a likeness that was already operating in the structure of pre-linguistic experience before it became visible as a tool. The novelist and the linguist both found it because it is genuinely there.

A figure, as Halliday and Matthiessen (2006) describe it, consists of three elements: a process, the participants involved in that process, and the circumstances attendant upon it. The process functions as the organizational centre — the element around which the others are arranged. Participants are directly involved in the process: they are the doers and done-to, the sensors and the said, the bearers of attributes. Circumstances are more peripherally attendant on the process: they supply the times, places, causes, and conditions within which the figure takes shape. As Halliday and Matthiessen put it, “participants are directly involved in the process; circumstances are more peripherally attendant on it” (2006: 172). The three elements together constitute a figure: a configuration perceived as a whole, prior to any grammaticalization that would distribute them across clausal slots. Participant roles are filled by things or qualities, circumstance roles by times, places, and causes, and the process role by a doing, happening, sensing, or being (Halliday and Matthiessen 2006: 58). What Halliday's account gives the present argument is not merely a vocabulary but a structural anatomy: the content plane is not an undifferentiated blur but an organized field, and it is organized in a specific way — as figures, each with its own internal articulation of process, participants, and circumstances — before any lexicogrammatical encoding arrives to capture it.

A third trajectory converges on the same point from within the critical literature on *The Inheritors* itself. Clark (2009), in a pragmatic stylistic analysis, argues that the difficulty readers experience with Neanderthal dialogue need not indicate its poverty: “we are missing out on something which they possess in their ability to communicate so effectively without words” (Clark 2009: 194). What looks like communicative deficit from inside the grammaticalized system may be richness at a level that system cannot register. Clark (2009) arrived at this conclusion through inferential pragmatics and theory of mind; the present paper arrives at it through stratification theory. The convergence is again independent, and again structural: the content plane is precisely the stratum that a reading confined to the inner strata will mistake for absence.

3. Halliday's Stratification and the Distance from Being

Halliday's model of language describes meaning as organized across multiple strata, each realizing the one above it in a nested hierarchy. As shown in Figure 2, moving outward from the innermost circle — phonology (sounding) — through lexicogrammar (wording) to semantics (meaning), each stratum represents a further layer of encoding between experience and expression. The content/expression division (Hjelmslev 1953), indicated by the dashed line, cuts across these strata, marking the boundary between meaning-making and its phonological realization.

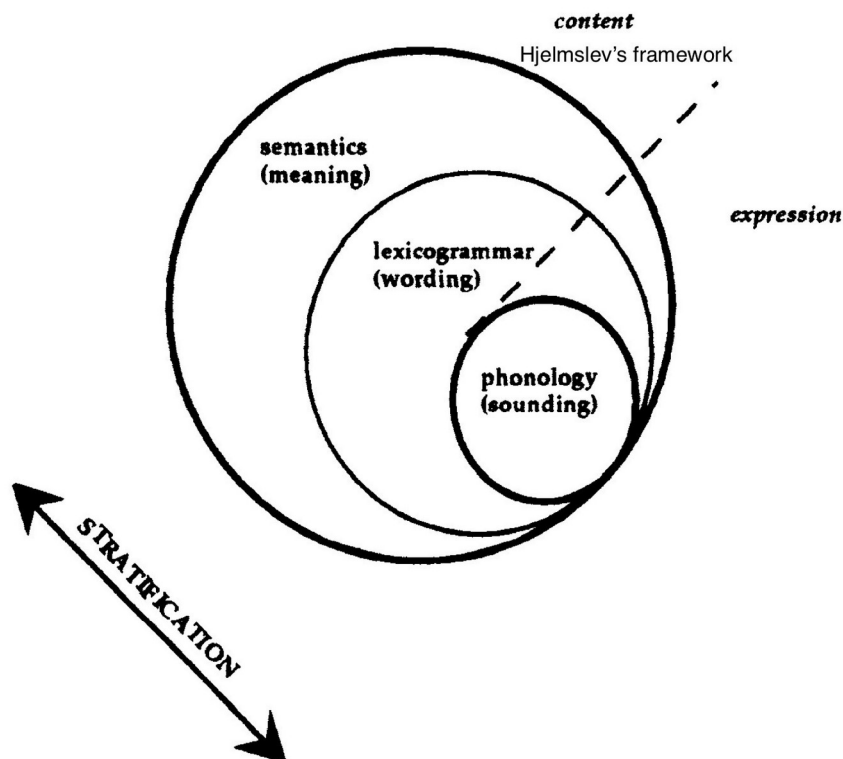


Fig. 2. Halliday's stratification model, showing the nested relationship between phonology, lexicogrammar, and semantics, with the content/expression division after Hjelmslev (1953). © M. A. K. Halliday and Christian M. I. M. Matthiessen, 1999, *Construing Experience Through Meaning*, Continuum, an imprint of Bloomsbury Publishing Plc. All rights reserved.

What the diagram does not explicitly show — but what Halliday's account implies — is the stratum that lies beyond the outermost circle: the content plane, the stratum of figures themselves, where experience is organized as configurations before semantics captures it. This is the stratum closest to immediate experience, the ground from which the entire stratified system grows. Each circle moving inward represents a further remove from that ground — a further encoding, a further layer of the negational apparatus that this paper associates with the (b) ↔ (c) loop and the progressive distance from (a1).

One terminological boundary deserves explicit notice, and one broader point it establishes. The content plane as used here designates the sub-propositional level: figures constituted by process, participants, and circumstances — the configurations themselves — before propositional framing turns them into potential facts that can be agreed with or denied. In Functional Discourse Grammar's terms, propositional content already belongs to the representational level, above states of affairs (SoAs) (Butler and Taverniers 2008). The content plane of this paper corresponds to that sub-propositional zone: SoAs and below in FDG's terms, not yet the propositional layer that enables truth-evaluation. The mapping with FDG is therefore strong and precise: it is at the SoA level and below — or in the Halliday diagram's terms, at the figural periphery beyond the outermost circle — that figures exist as configurations before semantic framing, and it is this level that Lok inhabits and that Halliday's figures describe. That this mapping holds across theoretically independent frameworks — FDG and SFG share no common architecture and arrive at equivalent levels through entirely different theoretical commitments — suggests that the content plane is not a

term of art specific to stratification theory but a structural level that independent approaches to linguistic organization keep identifying under different names. The convergences this paper assembles are not confined to philosophy, narrative, and cognitive science: they extend into the interior of linguistics itself.

The mapping onto the paper's framework is direct. The content plane, with its figures of participants, processes, and circumstances, corresponds to the closest approach to (a1) that language-using beings can achieve — the stratum where experience is already meaningful as configuration before the inner strata capture and encode it. Each stratum moving inward — lexicogrammatical, phonological — is another layer of mediation, another step of differentiation from primordial unity. The history the paper traces — from pre-Socratic immediacy through Platonism, modernity, Gestell, to AI — is simultaneously a history of progressive grammaticalization: Being encoding itself into ever more abstracted strata, moving further from the figural ground, until the innermost strata achieve such autonomy that they can run without human hands.

4. Lok at the Content Plane

Lok operates at the content plane. His picture-thinking is precisely figural in Halliday's sense: holistic configurations rather than sequential propositions, immediate rather than mediated by the negational apparatus, pre-grammatical in that they precede the encoding that would turn them into clauses and lexical items. When Lok encounters the log — as bridge, as canoe, as kindling — he is not selecting between multiple semantic interpretations of a single object. He is encountering different figures: different configurations of participant, process, and circumstance, each presenting itself as a unified whole in the circumstances of the encounter.

Golding's prose enacts this at the level of the clause. When the log is gone, Lok does not register an agent who moved it or a cause that explains its absence: he says, simply, "The log has gone away" (Golding 1955: ch. 1). The grammar gives the log itself the role of agent in its own departure — transitivity suppressed, causality unencoded, the configuration perceived as having withdrawn rather than been acted upon. This is not naïve syntax but precise figural rendering: a consciousness that encounters configurations as wholes has no grammatical apparatus for distributing them into the subject-verb-object structure that would turn a perceived absence into an explanatory proposition.

This raises a question the paper must address directly: if Lok operates at the figural periphery, outside the inner strata, why does he speak in clauses at all? The answer lies in Halliday's distinction between the content plane and the expression plane. Neanderthal speech in the novel is best understood as figural vocalization: the expression plane providing the only available medium for externalizing what is essentially a shared picture, without the full inner-strata semantic machinery behind it. When Lok says "It is bad to be alone," he is not producing a proposition in the full semantic sense — reasoning from premises, negating alternatives, encoding abstract categories. He is vocalizing a figure: a configuration of togetherness and apartness, bodily heaviness, immediate felt state. The grammatical form is the expression-plane vehicle; what it carries is content-plane substance. This accounts for the pervasive features of Neanderthal speech that Halliday (1971) identified and Hoover (1999) examined in detail — the present-tense immediacy, the intransitive verbs, the inanimate subjects, the suppressed causality — not as

syntactic deficits but as the natural shape of figural vocalization: the minimal grammatical form adequate to externalize a figure without passing it through the full semantic engine of the inner strata. It also explains how Mandala's (2017) finding of sophisticated inferential content in Neanderthal dialogue is compatible with content-plane cognition: the inferential richness is figural and embodied; what the inner strata add is not richness but abstraction.

What Lok lacks is not the content plane — that is precisely where he lives — but the semantic and lexicogrammatical strata closer to the center of the diagram. He cannot grammaticalize his figures into stable propositions, cannot lexicalize his processes into words that hold meaning across contexts, cannot abstract participants into categories that survive the immediate encounter. *Homo sapiens*, by contrast, operate primarily at the inner strata — so thoroughly that the content plane becomes largely inaccessible, overlaid by semantic and lexicogrammatical structures so internalized that experience arrives pre-grammaticalized before consciousness touches it.

What the log passage shows, and what the fungi-ears passage in chapter 10 will confirm, is not cognitive deficit but a different mode of meaning entirely — one operating at the stratum closest to immediate experience, which the inner strata simultaneously encode and progressively obscure.

5. The Fungi-Ears Convergence Revisited

The fungi-ears passage in chapter 10, where Lok discovers “like” — where the pre-predicative as-structure (in Heidegger's terms) achieves enunciation for the first time — can now be read with greater precision through the Halliday lens. At the content plane, fungi and ears can be the same figure: both are configurations of a certain shape, a certain softness, a certain spatial relationship to a surface. The as-structure Heidegger describes is operative here, beyond lexicogrammar toward the figural periphery, as a pre-linguistic figural resemblance. Fungi were ears not as a metaphor — metaphor requires the semantic stratum where two lexicalized meanings are mapped onto one another — but as a figural identity at the content plane. Same figure, different circumstances.

Golding marks this movement in the grammar of the passage itself. “Fungi on a tree were ears” (Golding 1955: ch. 10) — Golding's syntax gives us the copula of identity, not comparison: same figure, different circumstances, as the content plane presents it. The convulsion is then enacted in Lok's first direct speech: “The people are like a famished wolf in the hollow of a tree” (Golding 1955: ch. 10). *Like* has crossed from figural identity into predicated comparison — from a relation lived at the content plane to a semantic operation performed within it. The grammar of that sentence is the grammar of the threshold: not yet the fully grammaticalized inner strata, but likeness now functioning as a tool rather than presenting itself as a thing.

The convulsion in chapter 10 is then the moment this figural identity begins to be grammaticalized — begins to move inward through the strata toward something that could eventually become the semantic operation of metaphor and analogy. Lok does not quite reach full metaphor. He reaches the threshold — “likeness as a tool” (Golding 1955: ch. 10) — which is the moment the content plane figure begins to become a semantic resource. The convulsion is the movement between strata, not the arrival at the semantic level. What Golding dramatizes is a consciousness catching itself in the act of moving from one stratum to another — from the figural ground inward toward the encoded strata that *Homo sapiens* already inhabit entirely.

This reframes the companion paper's claim about the asymmetry between human and artificial intelligence. AI operates at the inner strata: it has the semantic and lexicogrammatical apparatus in full, with extraordinary sophistication. What it has never had is the content plane — the stratum of figures, of pre-grammaticalized experience, from which the inner strata grow. AI was given analogy as a semantic operation from the start, through training on millions of encoded examples. It was never at the content plane from which semantic operations emerge. The convulsion — the movement between strata — is not an experience AI can have, because it has never inhabited the stratum from which such movement begins.

6. The Shared Picture as Pre-Dissociative MAL

One further convergence is worth noting, though it requires the Kastrup (2019) framework from the companion paper to make visible (Figure 3). The Neanderthals do not merely think in figures individually — they share their pictures. “They shared a body as they shared a picture” (Golding 1955: ch. 5). The shared picture circulates between them as immediate intersubjective experience: the unperspectival “we” inhabiting a single figural field without requiring the individual perspectival “I” to translate it into language.

In Kastrup's (2019) framework, this shared figural field maps onto the pre-dissociative state of mind-at-large (MAL) — distributed, undivided consciousness before the Markov blanket individualizes it into bounded experience with private interiors. The shared picture at the content plane is what consciousness looks like before dissociation separates it into alters. The a2 human has fully private mental content — grammaticalized, lexicalized, encoded in individual interior language. The shared picture has been replaced by the individual proposition.

What a3 practice might recover — craftspeople working together in absorbed engagement, the carer and patient sharing a figural field before either articulates what they are perceiving, the teacher sensing the room's configuration before analyzing its dynamics — is something closer to the shared picture. Not the full pre-dissociative state, which is not available to beings who have passed through full representational consciousness, but figures becoming shareable again beyond the grammaticalized inner strata. The content plane as the stratum where individual interiors can briefly touch without full representational mediation. This may be, the Halliday convergence suggests, the most structurally precise account of what genuine contact with (a1) involves: not mystical dissolution but two or more practiced interiors sharing a figure before either of them turns it into a proposition.

The novel's closing image encodes this permeability structurally. Tuami sails away from the receding Neanderthal shoreline unable to stop sensing what is on the other side — the water flashes rather than darkens, dazzles rather than seals. In the Halliday framework, this is not merely a novelistic detail but a structural claim: the boundary between the content plane and the inner strata is not a hard edge but a permeable threshold, and it is permeable in both directions. Lok's convulsion moves outward-to-inward — a figure at the content plane beginning to be grammaticalized, likeness becoming a tool. Tuami's sensing moves inward-to-outward — a fully grammaticalized interior still being touched by what the inner strata have progressively enclosed. The membrane flickers; it does not seal. What a3 practice may recover is not the full pre-dissociative state — that is not available to beings who have passed through full representational consciousness — but the maintenance of that permeability from the inside: keeping the threshold

between content plane and inner strata flickering rather than allowing it to darken into an impermeable enclosure. Two practiced interiors working in absorbed engagement briefly inhabit the same figure before the inner strata individuate it into private propositional content — the craftspeople whose hands find the same grain before either names it, the carer who reads the body before articulating what they see. The Halliday framework gives this a structural name that neither phenomenology nor the Kastrup model alone can supply: shared access to the figural periphery, prior to the grammaticalization that would seal each figure inside a single bounded interior.

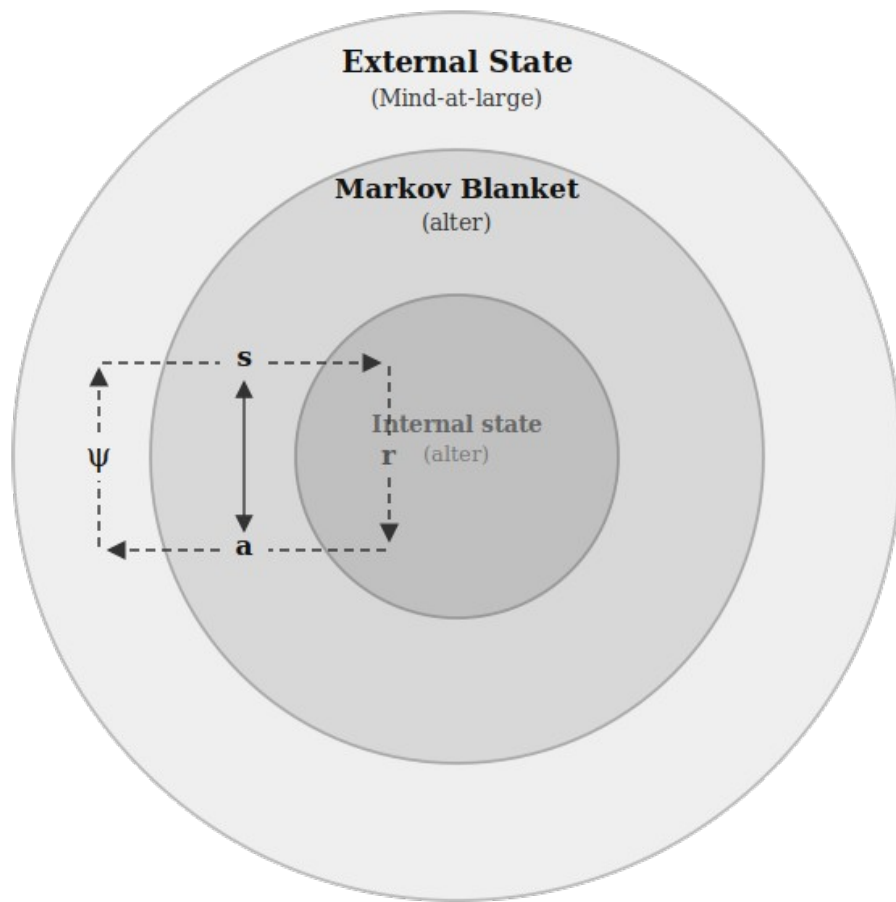


Fig. 3. Kastrup's dissociative boundary model, where MAL (mind-at-large) interacts with individual alters through a permeable Markov blanket of sensory states (s) and active states (a), with internal state (r) representing the alter's bounded experience. Adapted from Kastrup (2019).

7. The Deepest Implication

If the content plane is the stratum closest to (a1), and if the history of representation is a history of progressive grammaticalization — Being encoding itself into ever more abstracted strata — then the entire arc the paper traces acquires linguistic-structural precision. Pre-Socratic immediacy is close to the content plane: experience as figure, before the Platonic operation that would turn figure into form, immediate configuration into eternal idea. Platonism is the decisive grammaticalization:

the move from figure to semantic category, from configuration to concept defined through negation. Modernity intensifies the lexicogrammatical stratum: the subject-verb-object structure of Cartesian representation, the isolated “I” grammaticalized into the sovereign subject of a mechanical world-object. Gestell, for Heidegger (1993), is the moment the lexicogrammatical stratum runs autonomously — standing-reserve as the world fully grammaticalized into calculable resources. And AI is the moment the inner strata achieve complete autonomy from the figural ground: the semantic and lexicogrammatical apparatus running without any content plane beyond it, without any figures that were ever immediately experienced.

The a3 condition is then not merely a return to practice in Heidegger’s sense. It is the recovery of access to the content plane beneath the grammaticalized surface — the stratum where experience is already meaningful as configuration before the inner strata capture it. What practice does at a3 is thin the inner strata enough that the figural ground becomes accessible again — moving outward, back toward the periphery of the diagram. The craftsman feels the wood’s resistance before naming it: the figure is there before the semantic encoding arrives. The carer reads the body before articulating what they are seeing: the configuration presents itself before the lexicogrammatical apparatus captures it. What was lost in the progressive grammaticalization of experience — the pre-linguistic figural ground from which all inner strata grow — is what a3 practice begins, slowly and incompletely, to recover.

This account finds unexpected resonance in the enactivist tradition within cognitive science. Varela, Thompson, and Rosch (1991), in their foundational work *The Embodied Mind*, argue that cognition is not the manipulation of internal representations but a form of sense-making that arises from the dynamic coupling of organism and environment — what they call enaction. Meaning, on this account, is not encoded in advance and retrieved; it emerges in the act of engaged participation with a world that pushes back. Thompson’s subsequent development of this framework (2007) draws it explicitly into contact with phenomenology, arguing that the lived body — not the representational mind — is the primary site of meaning-making. This is structurally identical to the claim the Halliday framework supports: that the figural ground of experience is not produced by the inner strata but is always already there, the condition that the inner strata encode but cannot replace.

More precisely: what enactivism calls the sensorimotor loop — the continuous, pre-reflective coupling of perception and action through which the organism makes sense of its environment — operates at the content plane in Halliday’s terms. The loop produces figures: configurations of participants, processes, and circumstances that are meaningful before any lexicogrammatical structure captures them. When the enactivist says that the skilled practitioner does not represent the world and then act upon it but acts and thereby constitutes the world as meaningful, they are describing the same pre-grammaticalized ground that Halliday’s figures occupy and that Golding’s Neanderthals inhabit. The convergence extends across three independent traditions — phenomenological linguistics, narrative enactment, and embodied cognitive science — none of which was working from the others.

This is the deepest implication of the Halliday convergence. The paper’s claim that practice approaches (a1) where representation cannot is not merely a phenomenological or philosophical claim. It has a structural linguistic account: practice reactivates access to the content plane, the stratum of figures, the ground of meaning that language encodes but does not create and cannot replace. The convulsion — at any scale, individual or epochal — is always a movement between

strata. Something operating at the content plane suddenly becoming available at the semantic level. The figure becoming a resource. The ground becoming visible as having been generative all along.

8. A2 Convulsions: Analogies Made Inside the Strata

The account so far treats convulsion as a movement between strata — the Lok event, the pre-predicative as-structure achieving enunciation, the figural ground breaking inward into the semantic level. But something structurally distinct also happens within the inner strata, and it deserves its own account: the a2 convulsion, the analogical rupture that occurs entirely inside the encoded system.

When Gödel maps arithmetic onto meta-mathematics, when a physicist recognizes wave equations in population dynamics, when a poet discovers that grief and weather share a structure — something breaks open. These are not trivial pattern matches. They reorganize the semantic field. They produce meanings that neither domain contained before the mapping. In the strata diagram, these are lateral movements within the semantic circle: one region of encoded meaning suddenly illuminating another, the (b) ↔ (c) loop discovering that two of its own configurations are structurally identical. The convulsion is real. But it is contained. What breaks open is a representational framework, and what replaces it is another representational framework — more adequate, more capacious, but still fully grammaticalized. The new meaning was latent in the system's own encoded structure; the analogy surfaces what was always already there within the inner strata.

This explains a phenomenon that the companion paper gestures toward but does not fully account for: why a2 convulsions feel profound but do not transform. The scientist or mathematician who has a breakthrough — who sees the analogy that restructures the field — reports something phenomenologically close to what Lok experiences. The sudden visibility of what was already there. The sense of inevitability after the fact. But the transformation stays within the representational economy. The framework changes; the mode of being does not. The inner strata are reorganized but not thinned. The movement is lateral, not outward. Hofstadter's (2008) Gödel example is exactly this: meaning emerging from mapping between formal systems, entirely within the inner strata, never touching the figural periphery.

What distinguishes the Lok convulsion — and the a3 convulsion — is not the phenomenology of rupture, which may be similar in both cases, but the direction of movement. The a2 convulsion moves laterally within the inner strata: one semantic mapping illuminating another, one grammaticalized structure recognizing itself in a different domain. The a1-touching convulsion moves outward — toward the figural periphery, toward the content plane, toward what precedes the entire encoded system. The former produces new knowledge within the representational economy. The latter produces something the representational economy cannot fully absorb, which is why such encounters tend to be described in paradox, negation, and gesture rather than proposition. The Zen master's koan, Wittgenstein's ladder to be thrown away (Wittgenstein 1922: 6.54), Nāgārjuna's tetralemma (Nāgārjuna 1995) — these are not failures of articulation but accurate reports of a movement that the inner strata cannot encode without remainder.

The strata framework also gives structural precision to what the companion paper calls "representational colonization" — the risk at the a3 interface with AI that a sophisticated a2

convulsion substitutes for the slower, harder movement outward. AI is extraordinarily capable of generating lateral analogical mappings within the semantic stratum: mappings that feel rupturing and revelatory, that reorganize frameworks, that produce the phenomenology of breakthrough. The vast fluency of the field can simulate the experience of a1-contact without the underlying movement having occurred. In strata terms: the inner strata produce a sufficiently surprising lateral movement that the practiced interior mistakes it for outward movement toward the figural periphery. The phenomenology is similar. The direction is opposite.

The discrimination between these two movements may itself be the central practice of a3 existence — as the companion paper suggests, though without yet having the strata framework to name it precisely. A practiced interior calibrated through embodied engagement with the figural ground will have felt the difference between lateral reorganization and outward movement. Not because the outward movement is more dramatic — it may be quieter, less immediately productive of new encoded knowledge — but because it leaves a different residue. The lateral a2 convulsion leaves a new framework in place of the old one. The outward a1-touching convulsion leaves something that resists being framed at all: a sense of having been in contact with the ground from which all frames grow, which no subsequent frame fully captures. That remainder — the unspeakable residue of genuine outward movement — may be the most reliable marker of the difference.

9. Aging as a Second Forcing Condition: The Gerontolinguistic Evidence

The argument developed so far has traced a history of progressive grammaticalization ending in AI's completion of the representational project, and has identified a3 — the recovery of access to the content plane — as the condition that follows. The companion paper argues that this recovery is forced by AI displacement of representational work. But there is a second and universal forcing condition that the stratification framework allows to be named with greater precision than philosophy alone can supply: aging.

Every human being eventually faces reduced facility with the inner strata. The conventional account of cognitive aging is written almost entirely in deficit terms — processing speed declining, working memory narrowing, the lexicogrammatical apparatus becoming less fluid. Read through the Halliday framework, however, this pattern inverts its valence. If the inner strata are what progressively distance beings from the figural ground, their reduced facility is not only loss. Under certain conditions it is also structural return: the content plane becoming more accessible as the inner strata lose some of their totalizing grip. The gerontolinguistic literature, read through the stratification model, reveals a consistent empirical pattern that the deficit framing of that literature systematically misreads.

The most significant finding concerns what might be called the gist-over-microstructure shift. Older adults reliably process and retain the gist of discourse — what a text or conversation is about, its holistic situational shape — while showing relative decline in microstructural processing: the propositional detail network, the fully lexicogrammaticalized encoding of specific information (Radvansky et al. 2001). In Halliday's stratification model, the microstructure corresponds precisely to the inner strata: the propositional, lexicogrammatical encoding that constitutes the (b) ↔ (c) loop at its most developed. The gist — the holistic configuration of what is happening, who is involved, under what circumstances — maps onto what the content plane delivers before the

inner strata decompose it into propositional units. Older adults are not losing meaning. They are processing at a different stratum.

The most precise empirical support for this mapping comes from Chapman, Anand, Sparks, and Cullum (2006), who examined three distinct forms of gist in healthy seniors and mild Alzheimer's disease. Their central finding is that the three gist forms are not equivalent with respect to aging: they are affected in a gradient that runs from most to least dependent on propositional inner-strata processing. Main-idea gist — the recall of explicitly stated thematic units from a passage, requiring detailed propositional representation of content — declined significantly with advancing age. Transformed gist — defined as the generalised meaning of a passage derived by integrating world knowledge with what the text conveys, elicited as global synopsis and interpretive message — showed no significant difference between healthy young seniors (ages 65–79) and healthy old seniors (ages 80–95), despite the old senior group showing clear decline in detail memory across the same span. The dissociation is clean: detail memory falls, main-idea gist falls, transformed gist holds.

Categorical gist — the spontaneous clustering of words by semantic category during list recall — occupies an intermediate position in this gradient and deserves explicit account. It declines with healthy aging, but less severely than main-idea gist and detail memory. Chapman et al. note that this decline may reflect reduced spontaneous deployment of semantic categorization strategies rather than loss of the underlying capacity. In Halliday's terms, semantic categorization is itself an inner-strata operation: it requires the lexicogrammatical apparatus of abstract category recognition, the ability to impose semantic structure across items and deploy that structure deliberately during recall. Categorical gist is therefore closer to the inner strata than transformed gist, despite being a gist form rather than propositional detail. Its intermediate decline is consistent with the gradient: the more an operation depends on the inner-strata apparatus, the earlier and more severely aging affects it. The three gist forms, ordered by their distance from the content plane, age in precisely that order.

The gradient runs exactly as the Halliday framework predicts. Main-idea gist is closest to the inner strata: it requires selective attention to propositionally organised content, the identification of explicitly stated thematic units, the kind of operation that the (b) ↔ (c) loop performs at its most developed. Categorical gist sits at an intermediate distance: dependent on semantic categorization, which is an inner-strata operation, but less tightly bound to propositional detail than main-idea gist. Transformed gist is closest to the content plane: it is the holistic configuration of what an experience *means*, derived through integration with world knowledge rather than through propositional decomposition of what it *says*. It is figural in precisely Halliday's sense — a configuration perceived as a whole, prior to the grammaticalization that would distribute it into clausal slots. That it is this form — and not the more propositionally dependent forms — that remains robust into the ninth and tenth decades of life is not, on the Halliday account, surprising. It is the stratum the diagram places furthest out, closest to (a1), that aging reaches last.

Chapman et al. ground their findings theoretically in Brainerd and Reyna's Fuzzy Trace Theory (1990), which proposes that gist traces and verbatim traces are encoded independently and in parallel — gist-based memory is not simply abstracted upward from detail-based memory but runs alongside it on a separate track, fading more slowly and resisting decay more robustly. This theoretical independence of gist from detail is structurally consonant with what Halliday's stratification model implies: the content plane and the inner strata are distinct strata, not a hierarchy

in which gist is a compressed propositional residue. Both frameworks, arriving from entirely different disciplinary directions — cognitive neuropsychology on one side, systemic functional linguistics on the other — converge on the same structural claim: that the holistic, configurational, non-propositional processing of meaning operates independently of the inner-strata propositional apparatus, and that it is this independent operation that aging affects last. The convergence is, again, structurally independent. Neither Brainerd and Reyna nor Chapman et al. were working from Halliday. That the architecture they describe from within cognitive science maps so precisely onto the stratification model is itself evidence that the structure is real.

Crucially, Chapman et al. (2002, 2006) also find that transformed gist is the first form to collapse in mild Alzheimer's disease — where all three gist forms are impaired. This asymmetry between healthy aging and pathological aging is structurally significant for the present argument. What healthy aging does is reduce facility with the inner strata while leaving the content plane relatively intact: the stratum the diagram places furthest out is the last to be affected. What Alzheimer's disease does is collapse the content plane itself — destroying not just the propositional apparatus but the figural ground that the inner strata enclose. Despair in Erikson's (1950) sense, and mismanaged a3 more broadly, would occupy a middle position: reduced facility with the inner strata without access to the content plane having been cultivated through embodied practice, leaving the person stranded between strata — without the propositional resources they formerly relied on and without developed access to the figural periphery that remains. Alzheimer's is the structural worst case: both gone. Mismanaged a3 is the preventable case: reduced facility with the inner strata while the content plane, never developed as a resource, offers no figural ground to move outward into.

To the author's knowledge, no study has specifically tested Halliday's Neanderthal features — present-tense immediacy, intransitive verbs, inanimate subjects, suppressed causality — in the spontaneous speech of healthy older adults using stratification theory as its framework. Halliday (1971) identified these features as the grammatical shape of figural vocalization in Golding's prose; the gerontolinguistic literature has not yet approached aged speech from within that theoretical frame. What the existing literature does offer, however, is empirical data whose pattern is consistent with what stratification theory would predict: a systematic shift toward the stratum closest to immediate experience, away from the inner-strata operations of propositional encoding, abstract categorization, and embedded syntactic complexity. The gradient Chapman et al. document — transformed gist preserved, categorical gist intermediate, main-idea gist declining, detail memory declining further — is the empirical shape of what outward movement through the strata would look like if measured from inside the inner-strata framework that currently dominates gerontolinguistics.

The deficit framing is not wrong about what it measures. Processing speed declines. Working memory narrows. Complex syntactic structures become harder to sustain. These are real losses at the level of the inner strata. What the framing cannot see — because it has no concept of the stratum that lies beyond its outermost circle — is that the same reduced facility with the inner strata that registers as decline may simultaneously be opening access to what those strata have, across a lifetime, progressively enclosed. The gerontolinguistic literature is doing to aged speech exactly what pre-Halliday readings did to Golding's Neanderthals: measuring from inside the grammaticalized system, finding absence where there is a different kind of presence, calling figural richness communicative deficit.

Erikson (1950) made a distinction between integrity and despair in aging that can now be read with greater linguistic precision. What integrity preserves — the capacity to inhabit the life one has lived as a whole, without the representational scaffolding of future-orientation — is not merely an attitude. It is a mode of processing: the capacity to access the transformed gist, the holistic configurational meaning, when the propositional apparatus that formerly organized experience begins to lose its grip. Despair is not just the failure to accept aging. It is the failure to have developed, through embodied practice, access to the stratum that remains when the inner strata lose their facility. The person in despair is left without resources at the content plane — because the inner strata were the only strata they had learned to inhabit. What Chapman et al. document empirically — transformed gist preserved in healthy aging, collapsed in Alzheimer's — maps onto Erikson's distinction with unexpected precision: integrity is the healthy aging pattern, where the outermost stratum holds; despair approaches the Alzheimer's pattern structurally, where even the figural ground has been lost, whether to disease or to a lifetime of neglecting to cultivate access to the figural periphery.

What the convergences assembled across this paper collectively demonstrate is that the content plane is not a theoretical construct imposed on experience from above but a structure that independent traditions keep finding when they examine meaning from outside the inner-strata framework that normally dominates their own disciplines. Golding found it through a novelist's problem. Halliday found it through a linguist's problem. Varela, Thompson, and Rosch found it through a cognitive scientist's problem. Chapman, Anand, Sparks, and Cullum — working within cognitive neuropsychology, with no knowledge of stratification theory — found it through the empirical problem of distinguishing healthy aging from pathological aging. That the gerontolinguistic evidence arrives from the most distant disciplinary direction of all, and maps most precisely onto the stratification model's prediction about which stratum aging reaches last, may be its strongest recommendation. The content plane is not a conclusion the argument reaches. It is a structure the argument finds already waiting, in domain after domain, whenever the inner strata are examined from outside their own enclosure.

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